Differential Equations

Section 1: Separation of Variables

1 Solve the differential equation $\frac{dy}{dx} = \frac{2x}{y}$ with initial condition $y(0) = 4$.

a. $y^2 = 2x^2 + 16$

b. $y^2 = x^2 + 16$

c. $y = 2x + 4$

d. $y^2 = 2x^2 + 4$

2 Find the general solution of $\frac{dy}{dx} = xy$.

a. $y = Ae^{x^2/2}$

b. $y = Ae^{x^2}$

c. $y = \frac{1}{2}x^2 + C$

d. $y = Ae^x$

3 Solve the differential equation $\frac{dy}{dx} = 1 + y^2$.

a. $y = \tan(x + C)$

b. $y = \ln(1 + x) + C$

c. $y = e^x - 1$

d. $y = \arctan(x) + C$

Section 2: First-Order Equations Requiring Partial Fractions

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4 Find the general solution of $\frac{dy}{dx} = y(1-y)$.

a. $y = \frac{Ae^x}{1+Ae^x}$

b. $y = Ae^x - A$

c. $y = \ln|1-y| + x$

d. $y = \frac{1}{1+Ae^{-x}}$

5 Solve $\frac{dy}{dt} = k(M-y)$ with $y(0) = 0$ for constant $k, M > 0$.

a. $y = M(1-e^{-kt})$

b. $y = Me^{-kt}$

c. $y = M - e^{kt}$

d. $y = M(1+e^{-kt})$

6 A chemical reaction produces a substance according to $\frac{dx}{dt} = 2(4-x)^2$ with $x(0) = 0$. Find $x(t)$.

a. $x(t) = 4 - \frac{4}{8t+1}$

b. $x(t) = 4(1-e^{-2t})$

c. $x(t) = \frac{8t}{2t+1}$

d. $x(t) = 4 - \frac{1}{2t}$

Section 3: Rate of Change Modeling Problems

7 The rate of decay of a substance is proportional to its remaining mass m. Write down the differential equation governing $m(t)$ for constant $k > 0$.

a. $\frac{dm}{dt} = -km$

b. $\frac{dm}{dt} = km$

c. $\frac{dm}{dt} = -k$

d. $\frac{d^2m}{dt^2} = -km$

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8 Water leaks from a cylindrical tank of cross-sectional area A through a hole at the bottom according to Torricelli's Law: $\frac{dh}{dt} = -c\sqrt{h}$. If initial height is H, how long does it take to empty completely?

a. $T = \frac{2\sqrt{H}}{c}$

b. $T = \frac{\sqrt{H}}{c}$

c. $T = \frac{H}{c}$

d. $T = \frac{H^2}{2c}$

9 A population P grows according to the logistic model $\frac{dP}{dt} = 0.04P\left(1 - \frac{P}{2000}\right)$. What is the carrying capacity of the population?

a. 2000

b. 0.04

c. 1000

d. 50000

Section 4: Particular Solutions & Long-Term Behavior

10 For the logistic model in the previous question, at what population size P is the growth rate $\frac{dP}{dt}$ maximized?

a. $P = 1000$

b. $P = 2000$

c. $P = 500$

d. $P = 1500$

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11 Solve the differential equation $x\dfrac{dy}{dx} = y + 1$ with $y(1) = 3$.

a. $y = 4x - 1$

b. $y = 3x$

c. $y = 4x^2 - 1$

d. $y = 2x + 1$

12 A body cools according to $\dfrac{dT}{dt} = -0.1(T - 20)$. What does $T(t)$ approach as $t \to \infty$?

a. $20\,^\circ\text{C}$

b. $0\,^\circ\text{C}$

c. ∞

d. $-20\,^\circ\text{C}$

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